

TOCUMEN INTL, Panama

WMO: 787920

Lat: 9.071N

Lon: 79.383W

Elev: 135

StdP: 14.62

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	69.6	71.2	62.4	84.6	85.8	64.1	90.0	84.0	17.2	84.7	15.1	85.6	0.4	310	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	16.4	93.5	77.7	92.9	77.5	91.6	77.1	81.8	88.1	81.0	87.6	80.4	87.0	8.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	159.2	85.4	79.2	152.1	84.3	78.9	150.5	84.1	45.8	88.4	44.8	88.3	44.2	87.4	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.4	14.7	12.6	66.3	96.2	5.5	1.7	62.4	97.5	59.2	98.5	56.1	99.5	52.1	100.7		
			63.5	83.9	5.5	1.8	59.6	85.2	56.4	86.2	53.3	87.2	49.4	88.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	81.4	82.0	82.7	83.1	82.2	81.7	81.9	81.7	81.1	80.8	80.8	81.3	(6)
(7)		DBStd	2.01	1.76	1.76	1.94	1.82	2.16	1.87	2.11	1.97	1.77	1.72	1.86	1.83	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11582	974	895	1015	994	999	951	989	983	934	954	924	970	(10)
(11)		CDD65	6107	509	475	550	544	534	501	524	518	484	489	474	505	(11)
(12)		CDH74	60631	5236	5141	6235	6092	5455	4874	5108	4942	4417	4315	4023	4793	(12)
(13)		CDH80	22199	2289	2291	2841	2701	1906	1542	1714	1584	1262	1123	1151	1795	(13)
(14)	Wind	WSAvg	3.7	4.7	5.2	5.2	4.2	3.0	2.8	3.0	3.0	3.2	3.5	3.2	3.9	(14)
(15)	Precipitation	PrecAvg	96.8	1.3	0.5	0.8	3.9	12.0	11.3	10.1	10.7	11.8	14.6	12.9	6.9	(15)
(16)		PrecMax	150.4	6.6	2.1	2.7	12.5	18.7	21.1	15.6	16.9	19.4	22.8	25.9	21.8	(16)
(17)		PrecMin	74.6	0.2	0.2	0.0	0.1	5.2	4.9	5.4	6.0	6.4	10.6	8.1	1.6	(17)
(18)		PrecStd	15.8	1.5	0.5	0.8	2.7	3.0	3.8	2.4	2.3	3.3	3.3	4.3	5.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	93.5	94.8	95.2	93.5	93.2	93.4	93.3	91.7	91.4	91.6	92.5	(19)
(20)			MCWB	77.3	75.9	76.6	77.7	79.6	80.6	79.5	79.9	79.7	80.2	79.8	78.9	(20)
(21)		2%	DB	91.5	91.7	93.1	93.4	91.6	91.0	91.4	91.2	89.6	89.1	89.7	91.1	(21)
(22)			MCWB	75.8	75.2	75.7	77.2	79.0	79.8	79.4	79.9	79.2	79.5	78.7	78.0	(22)
(23)		5%	DB	90.0	91.1	91.6	91.7	89.9	89.1	89.6	89.3	87.9	87.4	87.9	89.7	(23)
(24)			MCWB	75.3	75.1	75.3	76.6	78.7	79.4	79.2	79.6	79.2	79.1	78.7	77.2	(24)
(25)		10%	DB	89.4	89.7	90.4	90.8	88.0	87.4	87.8	87.5	86.1	85.7	86.1	88.1	(25)
(26)			MCWB	75.2	74.8	75.0	76.4	78.8	79.1	79.2	79.2	78.6	78.4	78.2	77.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	79.6	80.9	81.9	82.4	81.8	82.1	83.0	81.8	81.7	81.4	81.4	(27)
(28)			MCD B	89.2	88.3	89.5	89.4	88.1	89.1	87.7	88.3	87.1	87.5	88.1	88.6	(28)
(29)		2%	WB	78.6	78.2	78.6	80.0	81.0	80.8	80.8	81.2	80.4	80.3	80.1	80.0	(29)
(30)			MCD B	87.7	87.3	87.9	87.9	87.5	87.8	87.5	87.3	86.3	85.9	86.4	87.1	(30)
(31)		5%	WB	77.5	77.1	77.4	79.1	80.1	80.0	79.9	80.1	79.6	79.4	79.3	79.1	(31)
(32)			MCD B	86.9	87.0	87.8	86.9	86.6	86.4	86.5	86.3	85.1	84.4	85.0	85.9	(32)
(33)		10%	WB	76.5	76.2	76.4	78.3	79.2	79.2	79.2	79.2	79.0	78.9	78.7	78.4	(33)
(34)			MCD B	86.0	86.3	87.1	86.2	85.5	85.0	85.5	85.1	84.3	83.5	84.1	84.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.4	17.4	17.2	16.4	12.7	12.0	12.7	12.6	11.7	11.1	12.2	14.7	(35)
(36)			MCDBR	18.3	18.8	18.6	18.4	15.4	14.6	15.1	15.1	14.0	13.7	14.7	16.7	(36)
(37)			MCWBR	6.7	6.3	6.2	6.6	6.3	6.4	6.6	6.8	6.4	6.1	6.5	6.8	(37)
(38)		5% WB	MCDBR	17.5	17.1	17.4	16.0	13.3	13.1	13.4	13.7	12.5	12.1	13.1	14.8	(38)
(39)			MCWBR	7.7	7.0	7.3	7.1	6.5	6.5	6.6	6.9	6.2	5.9	6.4	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.421	0.448	0.464	0.481	0.473	0.444	0.455	0.456	0.428	0.412	0.434	0.422	(40)	
(41)		taud	2.412	2.322	2.283	2.240	2.302	2.389	2.363	2.358	2.427	2.497	2.422	2.440	(41)	
(42)		Ebn at Noon	282	278	274	266	262	267	265	269	280	285	276	279	(42)	
(43)		Edh at Noon	37	42	44	46	42	38	39	40	38	35	37	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1628	1751	1756	1707	1439	1331	1317	1355	1384	1354	1253	1420	(44)	
(45)		RadStd	74	88	97	79	107	88	86	74	78	69	104	152	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (1 neighbor)	N/A	N/A	-0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page